Part B

B6

Answer:- $T(X) = \left(\sum_{i=1}^{n} \ln(X_i), \sum_{i=1}^{n} \ln(1-X_i)\right)$

For $X_1, \ldots, X_n \overset{iid}{\sim} Beta(\alpha, \beta)$ with $f(x;\alpha,\beta) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}$

$\alpha, \beta > 0$, $0 \leq x \leq 1$

$\therefore$ Likelihood $L(\alpha,\beta|x) = \prod_{i=1}^{n} f(x_i;\alpha,\beta) = \prod_{i=1}^{n}\left[\frac{1}{B(\alpha,\beta)} x_i^{\alpha-1}(1-x_i)^{\beta-1}\right]$

After Factorizing $L(\alpha,\beta|x) = \left(\frac{1}{B(\alpha,\beta)}\right)^n \cdot \left[\prod_{i=1}^{n} x_i^{\alpha-1}\right] \cdot \left[\prod_{i=1}^{n}(1-x_i)^{\beta-1}\right]$

Now $\prod_{i=1}^{n} x_i^{\alpha-1} = exp\left(\ln\left(\prod_{i=1}^{n} x_i^{\alpha-1}\right)\right) = exp\left((\alpha-1)\sum_{i=1}^{n}\ln(x_i)\right)$

And $\prod_{i=1}^{n}(1-x_i)^{\beta-1} = exp\left((\beta-1)\sum_{i=1}^{n}\ln(1-x_i)\right)$

$\therefore L(\alpha,\beta|x) = \left(\frac{1}{B(\alpha,\beta)}\right)^n \cdot exp\left((\alpha-1)\sum_{i=1}^{n}\ln(x_i) + (\beta-1)\sum_{i=1}^{n}\ln(1-x_i)\right)$

$$= \underbrace{\left(\frac{1}{B(\alpha,\beta)}\right)^n exp\left(\alpha\sum_{i=1}^{n}\ln(x_i) + \beta\sum_{i=1}^{n}\ln(1-x_i)\right)}_{g(T(X);\alpha,\beta)} \cdot \underbrace{exp\left(-\sum_{i=1}^{n}\ln(x_i) - \sum_{i=1}^{n}\ln(1-x_i)\right)}_{h(x)}$$

g function depends on parameters α & β & involves the sample data only through 2 statistics

$T_1(X) = \sum_{i=1}^{n}\ln(X_i)$ $T_2(X) = \sum_{i=1}^{n}\ln(1-X_i)$

h function depends on data X But NOT on parameters α, β

$\therefore$ By NeyMann Fisher Factorization theorem the 2dimentional sufficient statistic is:-

$$\boxed{T(X) = \left(\sum_{i=1}^{n}\ln(X_i), \sum_{i=1}^{n}\ln(1-X_i)\right)}$$

any one-to-one transform of this pair, e.g. $\left(\prod X_i, \prod(1-X_i)\right)$ is also sufficient.